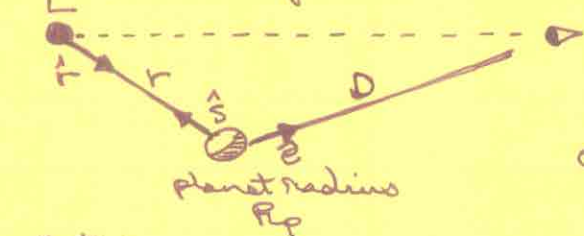


## Reflected Flux

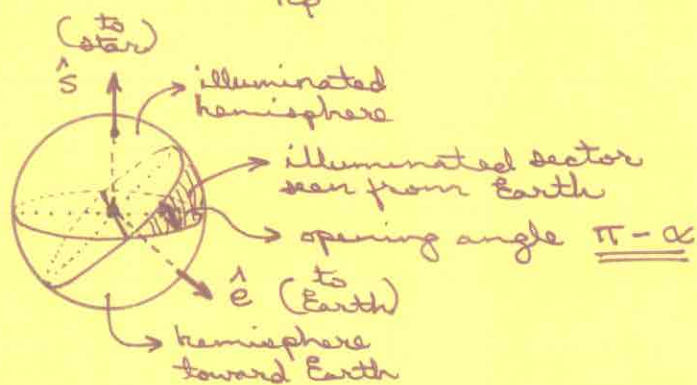
stellar luminosity



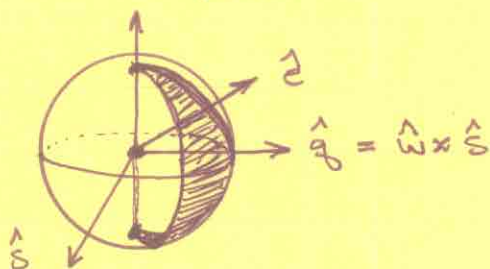
net power received by planet

$$= \frac{L}{4\pi r^2} \times \pi R_p^2 = L \left( \frac{R_p}{2r} \right)^2$$

$$\cos \alpha = \hat{s} \cdot \hat{e}$$



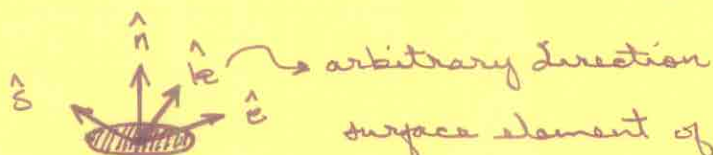
$$\hat{w} = \frac{\hat{s} \times \hat{e}}{|\hat{s} \times \hat{e}|}$$



Let  $(\theta, \phi)$  be coordinates on the sphere, such that the surface normal is

$$\hat{n} = \sin \theta \cos \phi \hat{s} + \sin \theta \sin \phi \hat{q} + \cos \theta \hat{w}$$

integral over sector :  $\int_{\alpha - \pi/2}^{\pi/2} d\phi \int_0^{\pi} d\theta \sin \theta$  (net solid angle)  $= 2(\pi - \alpha)$



surface element of area  $dA = R_p^2 d\phi d\theta \sin \theta$

power intercepted by element  $dP = \frac{L}{4\pi r^2} dA \hat{n} \cdot \hat{s}$

$$\hat{n} \cdot \hat{s} = \sin \theta \cos \phi$$

- let the total reflected power from the element be  $f dP$  per unit solid angle
- let  $f dP f(\hat{k})$  be the reflected power, such that  $\int_{\text{hemisphere}} f dP f(\hat{k}) = 1$ . Lambert's assumption is that  $f(\hat{k}) = f_0 \hat{n} \cdot \hat{k}$

$\Rightarrow f_0 = \frac{1}{\pi} \Rightarrow$  reflected power per solid angle toward Earth :  $f dP \frac{1}{\pi} \hat{n} \cdot \hat{e}$

⇒ contribution to flux at Earth from  $dA$  is

$$\underline{dF_e = \gamma dP \frac{1}{\pi} \hat{n} \cdot \hat{e} \frac{1}{D^2}} \quad ; \quad \hat{e} = s\alpha \hat{s} + s\phi \hat{q}$$

$$\Rightarrow \hat{n} \cdot \hat{e} = s\theta [c\alpha c\phi + s\alpha s\phi]$$

$$= \underline{s\theta c(\alpha - \phi)}$$

Integration over the observable illuminated sector gives the total flux at the Earth:

$$F_e(\alpha) = \boxed{\gamma} \frac{L}{D^2} \frac{1}{4\pi r^2} R_p^2 \frac{1}{\pi} \underbrace{\int_{\alpha-\pi/2}^{\pi/2} d\phi}_{\frac{dA}{R_p^2}} \underbrace{\int_0^\pi d\theta s\theta}_{\hat{n} \cdot \hat{s}} \underbrace{s\theta c\phi}_{\hat{n} \cdot \hat{e}} s\theta c(\alpha - \phi)$$

angular integral:  $\int_{\alpha-\pi/2}^{\pi/2} d\phi c\phi c(\alpha - \phi) \int_0^\pi d\theta s^3\theta$

$$c\phi c(\alpha - \phi) = \frac{1}{2} [c(\phi + \alpha - \phi) + c(\phi - \alpha + \phi)] = \frac{1}{2} [c\alpha + c(2\phi - \alpha)]$$

$$\frac{1}{2} \int_{\alpha-\pi/2}^{\pi/2} d\phi [c\alpha + c(2\phi - \alpha)] = \frac{1}{2} (\pi - \alpha) c\alpha + \frac{1}{2} s(\pi - \alpha)$$

$$s(\pi - \alpha) = s\alpha \Rightarrow \frac{1}{2} [s\alpha + (\pi - \alpha) c\alpha]$$

$$\int_0^\pi d\theta s^3\theta = \int_{-1}^1 du (1 - u^2) = u - \frac{1}{3} u^3 \Big|_{-1}^1 = 2 \left(1 - \frac{1}{3}\right) = \frac{2}{3} \cdot 2 = \frac{4}{3}$$

$$\Rightarrow \text{ang. int.} = \frac{2}{3} [s\alpha + (\pi - \alpha) c\alpha]$$

$$\text{let } \Phi(\alpha) = \frac{1}{\pi} [\delta \alpha + (\pi - \alpha) c \alpha] \quad \in (0, 1)$$

$$\Rightarrow F_e(\alpha) = \delta \frac{L}{4\pi r^2} \frac{R_p^2}{D^2} \frac{2}{3} \Phi(\alpha)$$

$$= \delta \underbrace{\frac{L}{4\pi D^2}}_{F_*} \frac{R_p^2}{r^2} \frac{2}{3} \Phi(\alpha)$$

$F_* \rightarrow$  flux from star

$$\Rightarrow \boxed{\frac{F_e(\alpha)}{F_*} = \delta \frac{2}{3} \left(\frac{R_p}{r}\right)^2 \Phi(\alpha)}$$

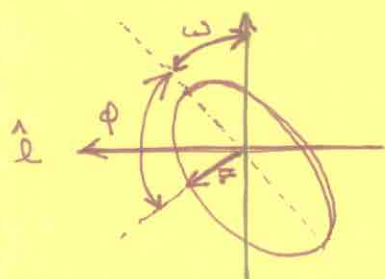
The orbit of the planet is described by  $\vec{r}(t) = r(t) \hat{r}(t)$ .

We have

$$\cos \alpha = \hat{s} \cdot \hat{e} = (-\hat{r}) \cdot (-\hat{l}) \quad ; \quad \hat{l} = \text{unit vector from Earth to star} \\ = \hat{r} \cdot \hat{l}$$

In terms of the usual orbital angles:

$$\hat{r} \cdot \hat{l} = s(\omega + f) s i \quad ; \quad \begin{array}{l} \omega = \text{argument of pericenter} \\ f = \text{true anomaly} \\ i = \text{inclination} \end{array} \\ (i = 90^\circ)$$



$$\Phi(\alpha) = \frac{1}{\pi} [s\alpha + (\pi - \alpha)c\alpha]$$

$$c\alpha = s(\omega + f) s i$$

$$s\alpha = [1 - c^2\alpha]^{1/2} > 0 \quad (\alpha \in [0, \pi])$$

$$\Rightarrow \alpha = \underbrace{\text{atan2}}_{\text{FORTRAN intrinsic}}(s\alpha, c\alpha)$$

FORTRAN intrinsic